PairUp

BUSINESS CASE · OCTOBER 2026

Say what you need. A neighbor has it.

A C2C marketplace on Alexa+ that pairs people who need something with people who have it or can do it, at a fair price set by a published rule.

01

Faster and cheaper to buy what you need

One sentence finds it. The whole price goes to the other person.

02

Make personal business great

Anyone opens a small business in one sentence and keeps what they earn.

03

C2C in the future

The open meeting point where personal AI agents trade for people.

Launch market: University of Washington, Seattle

Amazon Developer Hackathon · Alexa+ track

EXECUTIVE SUMMARY

Small local trades are worth billions, and most of them never happen

The people who need something and the people who have it are often a short walk apart. Finding each other, agreeing a price and paying a platform's fee cost more than a $15 trade is worth, so the trade is skipped.

$3–4K
of resellable items sit unused in the average US home
eBay Recommerce Report 2026[1]

13–20%
of each sale kept by major resale sites
Poshmark, Mercari, eBay fee schedules[2]

72.9M
Americans already earn money working for themselves
MBO Partners 2025[3]

600M+
Alexa devices sold; 97% can run Alexa+
Amazon at CES 2026[4]

The opportunity

PairUp lets anyone say what they need, have or can do. It pairs the two sides, settles the price inside limits each side set privately, proposes a public hand-off point and asks both people for a yes. Things, errands, hourly help and event staffing all run through the same engine, by voice.

Why it wins

- **Cheaper for both sides.** The agreed price passes in full from buyer to seller. On a $20 calculator, the seller receives $18.00 on PairUp against $16.00–17.12 on resale sites, and the buyer pays $18.00 against $20 plus shipping.
- **Faster.** One sentence replaces photos, a listing form, chat and haggling. Requests stand, so the next matching seller is paired the moment they post.
- **Built for agents.** Any AI assistant connects over the open MCP standard. Prices come from a published, open-source rule, so agents never haggle and cannot be talked into a worse deal. Research shows AI negotiators can be swayed by tone.[15]
- **Strong fit with Amazon.** Alexa+ is the voice, Amazon Hub Lockers and Counters are the hand-off points, amazon.com covers every need no neighbor can meet, and AWS runs it.

> **Recommendation.** Launch at the University of Washington in Seattle: 52,316 students in walking distance with needs that repeat every quarter. Our model puts one campus at about **$1.9 million a year** of trades between students. Measure for one quarter, then expand campus by campus.

THE PROBLEM

Trading with the person next door is slow, costly and risky

Five pains, each backed by public data.

PAIN	EVIDENCE
Fees eat small sales	Poshmark keeps 20% of sales of $15 and up; Mercari 10% plus 2.9% + $0.30 for payment; eBay about 12.9–15% plus $0.30 per order.[2]
Small errands cost more than the errand	A $39 restaurant meal cost $47.10 on the major delivery apps before service fees.[5] Grocery items averaged 11% above in-store prices, plus delivery fees.[6] Hired help adds a service fee and a Trust & Support fee on each hour.[7]
Strangers are risky	Americans reported $12.5 billion lost to fraud in 2024, up 25%. Online shopping was the second most reported category, with $432 million lost.[8]
Haggling leaks and stalls	On millions of eBay Best Offer negotiations, the listing price itself signalled how eager a seller was: round-number listings drew offers 8–12% lower.[13] Bargaining with private values loses deals both sides wanted.[12]
Useful things are thrown away	The US generated 12.1 million tons of furniture waste in 2018; 80.1% went to landfill.[9] Students are budgeted about $1,290 a year for books and supplies.[10]

Why the market skips these trades

Economics has explained this for decades. Searching costs time, so buyers settle for worse prices (Stigler, 1961).[11] Every market trade carries costs of finding, negotiating and enforcing it, and trades worth less than those costs never happen (Coase, 1937).[11] A $4 lunch run and a $15 lamp sit squarely below that line today.

> **What this means.** The demand and the supply already exist on every campus and street. The missing piece is a way to connect them that costs one sentence and charges no commission on the deal.

THE SOLUTION

One sentence in, one yes out

PairUp runs as an MCP server for Alexa+ and any AI assistant. People talk; code pairs, prices and schedules.

SELLER

"Alexa, sell my TI-84 for $20. I'll take anything above $15."

BUYER

"Alexa, I need a graphing calculator today, max $18."

ALEXA+

"Found it: TI-84 Plus for $18, $2 below asking. Meet-up at the Amazon Hub Counter on campus. Should I confirm it?"

1

Say it

Need, offer or skill, by voice or text. Requests stand until a match appears.

2

Pair

Semantic search, distance, urgency and track record rank every fitting offer.

3

Price

One published rule settles the price inside both private limits.

4

Yes

Both people confirm. They meet at a public point and pay at the hand-off.

Six markets, one engine

MARKET	EXAMPLE
Used things	Textbooks, calculators, lamps, a mini fridge at move-out
Errands	Lunch from the HUB to McCarty Hall, $4; cold medicine to a dorm
Hands by the hour	Two hours of moving help with a trolley, $12 an hour
Organizations	A club staffs registration, AV and a photographer for one Saturday
Neighborhood care	Cat sitting and plant watering while away
Needs not yet spoken	The usual weekday lunch run, offered before noon; every post waits for a yes

Trust built in

Private limits never leave the server. A price check asks "did you mean…?" on prices far from normal and warns buyers about deals that look too good. Banned items and services are refused with a reason. Ratings build each person's track record, reports pause pairing until reviewed, and hand-offs happen at public Amazon Hub points.

VALUE · FASTER AND CHEAPER TO BUY WHAT YOU NEED

The same calculator, four ways

A $20 TI-84. The buyer's budget is $18; the seller's private minimum is $15.



	SELLER RECEIVES	BUYER PAYS	EFFORT FOR THE SELLER
Poshmark (20%)	$16.00	$20 + shipping	Photos, listing, offers, shipping label
eBay (about 13.25% + $0.30)	$17.05	$20 + shipping	Photos, listing, offers, shipping
Mercari (10% + 2.9% + $0.30)	$17.12	$20 + shipping	Photos, listing, offers, shipping
PairUp	**$18.00**	**$18.00**	**One sentence and one yes; same-day hand-off on campus**

Fee rates from published fee guides;[2] resale rows assume the item sells at its $20 ask.

For errands

The lunch costs its counter price. The requester offers up to $5, a student already walking that way takes it for $4, and the whole $4 goes to the student who carried it.

For the buyer's time

No browsing, no shipping wait. A standing request keeps working: when the next matching seller posts, PairUp pairs them at once and tells the buyer's agent.

VALUE · MAKE PERSONAL BUSINESS GREAT

Everyone has a small business in them

A desk to sell in June, a free afternoon, a trolley, a daily walk past the pharmacy. PairUp turns each into income with one sentence.

72.9M
US independent workers in 2025, including 37.4M occasional earners
MBO Partners[3]

30%
of US adults aged 18–29 have earned money through an online gig platform
Pew Research Center[17]

2×+
the surplus drivers get from choosing their own hours, against fixed schedules
Chen et al., JPE 2019[18]

A SELLER ON PAIRUP	HOW
Opens in one sentence	"I do lunch runs from the HUB on weekdays, $4." The offer stands and is paired whenever a request fits.
Sets the price	A public ask, a private minimum, and automatic acceptance of any offer at or above it.
Keeps the earnings	The agreed price is paid in full at the hand-off.
Sets the hours	PairUp books work only inside the hours the provider states, and one person's bookings never overlap.
Builds a reputation	Ratings after every deal feed a track record that moves reliable people up the ranking.
Gets regulars	PairUp learns routines like the weekday lunch run and offers them again, so a regular helper gets regular work.

From research. Peer markets let owners earn from goods they use only part of the time (Horton & Zeckhauser, 2016),[19] and add the most value at peak moments when regular supply runs short (Farronato & Fradkin, AER 2022)[20]: the exam-morning calculator, the move-out van.

MARKET OPPORTUNITY

A large market, entered one campus at a time

Secondhand goods, independent work and voice assistants are each large and growing. PairUp sits where they meet.

LAYER	SIZE
US secondhand apparel (one category)	Grew 14% in 2024; projected $74 billion by 2029. 58% of US consumers bought secondhand apparel in 2024.[21]
Idle goods in homes	About $3,000–4,000 of resellable items per household.[1]
People earning on their own	72.9 million US independent workers.[3]
Voice reach	Over 600 million Alexa devices sold; 97% can run Alexa+.[4]
US college students	About 19 million; fall 2025 enrollment up 1.0%.[22]
Launch campus	52,316 students at UW Seattle, autumn 2025.[23]

One campus, modelled

Inputs (change any and recompute)

- 52,316 UW Seattle students
- 10% use PairUp in a quarter: about 5,200 people
- 2 exchanges a month each
- $15 average: a textbook, a calculator, a lamp or two lunch runs

10,500
exchanges a month

$1.9M
a year between students

$250–375K
a year kept from 13–20% commissions

$680M
a year at the same rates across US students

Model figures are our estimates from the inputs shown. The UW pilot will replace them with measured results.

Why campuses first

Campuses are dense and walkable. Needs repeat every quarter (textbooks, calculators), and move-in and move-out refill supply every year. Student clubs bring organization demand. Each new campus is a self-contained market that reaches density quickly.

BUSINESS MODEL

How PairUp earns

In the pilot the seller keeps the full price, ranking has no paid placement, and every deal is free. The streams below are future options, sized on the same UW inputs.

125,600
deals a year on one campus (model)
5,232 users × 2 deals × 12 months

$1.9M
traded a year (model)
125,600 deals × $15 average

$36–54K
revenue a year per campus (model)
Before and after the later seller fee

STREAM	INPUTS	PER CAMPUS A YEAR
Community license University, housing office or apartment operator	Flat fee, assumed $25,000 a year	~$25,000 (model)
Protected online payment Buyer pays a flat $1	30% of deals use it: 37,670 deals. Card processing on $15 is 2.9% + $0.30 = $0.74, leaving $0.26 a deal	~$10,000 (model)
Later: seller fee 3% on protected-payment deals only; cash deals stay free	37,670 deals × $15 × 3%	~$17,000 (model)
Amazon referrals Requests nobody can fill	Small share of unmet requests buying on amazon.com	~$1,000–2,000 (model)
Agent access tier Commercial MCP agents and apps	Volume pricing	Upside, not counted
Total	**Sum of the rows above**	**~$36,000 before the seller fee; ~$54,000 with it (model)**

At scale. 100 campuses and communities bring about $3.6–5.4 million a year (model). Costs stay small: one Graviton EC2 server, Bedrock billed per call, and no hand-off staff, because Ring presence at PairUp Points corroborates each hand-off. Each new community adds revenue at little added cost.

Model figures are our estimates from the inputs shown. The UW pilot will replace them with measured results.

WHY NOW · C2C IN THE FUTURE

Personal agents will trade for people. They need a fair place to meet.

Commerce is moving from apps people browse to assistants that act for them. Research on AI agents in markets points to three requirements, and PairUp meets each one.

WHAT RESEARCH FINDS	SOURCE	PAIRUP'S ANSWER
AI negotiators show anchoring; one acting desperate gains about 20%	Bianchi et al., ICML 2024[15]	Agents never bargain. One published rule in open-source pairing-core sets every price.
AI shoppers favor the first offer they receive, and quality falls as markets grow	Magentic Marketplace, Microsoft Research 2025[16]	The server ranks every fitting offer before an agent sees it, so the best match wins.
Markets of AI agents need new institutions to work well	Hadfield & Koh, NBER Handbook 2025[24]	An open protocol (MCP), shared test vectors for the price rule, and a required human yes.
No bilateral trading rule is perfect on every count	Myerson & Satterthwaite 1983[12]	A simple rule with its trade-offs published: public ask, automatic best-offer acceptance.

Open by design

- **Any assistant connects:** Alexa+, a Strands agent on Bedrock AgentCore, or any MCP client, through OAuth 2.1 with Login with Amazon.
- **Any host settles the same way:** pairing-core is MIT-licensed Rust compiled to WebAssembly, with shared test vectors.
- **People stay in charge:** agents find, explain and prepare; people confirm. Every person can be a buyer and a seller at once, through the assistant they already use.

Fit with Amazon

Alexa+
The voice for local buying and selling

Amazon Hub
Lockers and Counters as safe hand-off points

amazon.com
The new price and a link when no neighbor has it

AWS
Bedrock, AgentCore, CloudFront, EC2, CloudWatch

POSITION

One front door for needs that live in five apps today

TODAY'S OPTION	THINGS	ERRANDS	HOURLY HELP	PRICE SET BY	SELLER PAYS
Group chats and forums	●	◐	◐	Haggling in chat	Nothing
Resale sites	●	–	–	Haggling or offers	13–20% of the sale
Delivery apps	–	◐ food, groceries	–	The platform	Commission; buyer pays fees
Task apps	–	●	●	Hourly rate	Client pays service fees
PairUp	●	●	●	**Published rule, private limits**	**Nothing on the deal**

● full · ◐ partial · – not offered. PairUp also takes voice and AI-agent requests, keeps requests standing, and learns routines.

Risks and how we handle them

RISK	PLAN
Too few people at first	Start on one dense campus; requests stand until a match arrives; move-in, move-out and club events bring both sides at once.
Safety between strangers	Signed-in accounts, ratings, reports that pause pairing, banned-item rules, a too-good-to-be-true warning, and public Amazon Hub meet-ups.
Agents gaming prices	Prices come from code; one offer below the ask per buyer and listing; limits never leave the server; a human yes on every deal.
Mistyped or misheard prices	The price check asks "did you mean…?" against reference prices and the market's own trades, weighted so staged trades barely move it.
Payment	People pay each other at the hand-off today; in-app payment through Amazon is the natural next step.

How we will measure success

PROMISE	METRIC
Faster	Time from request to proposed deal
Cheaper	Price paid against the typical range and the new price on Amazon
Personal business	Share of the price kept by sellers; repeat providers; returning routines
C2C future	Requests paired from either side and through outside agents
Trust	Unaided task success; correct understanding of cost and time; reports per thousand deals

ROADMAP AND RECOMMENDATION

Start at UW. Grow campus by campus.

NOW	NEXT	THEN	BEYOND
UW Seattle	**Seattle area**	**US campuses**	**Every neighborhood**
Campus and nearby neighborhoods; 52,316 students	Other campuses and neighborhoods	About 19 million students, with campus Hub points	Any household, any MCP-connected agent

Ready today

- A live MCP server (spec 2025-11-25) with OAuth 2.1, MCP Apps cards, and a web app that plays the Alexa+ conversation in any browser.
- Pairing from either side with pgvector and full-text search; prices by open-source pairing-core; schedules checked against each helper's hours.
- Trust and safety: banned-item rules, ratings, reports, price check, data deletion.
- Learned routines, Bee context and an Apple Watch view; every suggestion waits for a yes.
- Running on AWS with Bedrock, a Strands agent for AgentCore Runtime, CloudFront, EC2 Graviton and CloudWatch.

Recommendation

1. **Launch the UW Seattle pilot** with students on both sides: requesters and providers.
2. **Measure for one quarter**: time to deal, price against alternatives, seller earnings, unaided success.
3. **Connect Amazon services**: the Hub locator for hand-offs and live amazon.com prices for the fallback.
4. **Expand** to the next Seattle campus once the first reaches steady weekly trades.

Say what you need. A neighbor has it. PairUp makes the deal fair, safe and fast.

EVIDENCE

References

Full notes on each paper: LITERATURE.md. All figures with context: IMPACT.md.

1. eBay Recommerce Report 2026, reported in La Voce di New York, Oct 9, 2026.
2. Seller fee guides for Poshmark, Mercari and eBay, Closo, 2025.
3. MBO Partners, State of Independence 2025.
4. TechCrunch, Amazon says 97% of its devices can support Alexa+, Jan 12, 2026.
5. Star Tribune, How to spot hidden charges in food delivery apps.
6. CHOICE, DoorDash, Milkrun and Uber Eats grocery prices compared.
7. Taskrabbit Help Center, What's the Taskrabbit Service Fee?
8. FTC Consumer Sentinel 2024 data, reported by SecurityWeek, March 2025.
9. US EPA, Durable Goods: Product-Specific Data (2018 figures).
10. College Board figures via BestColleges, 2024–25.
11. Stigler, G. J. (1961). The Economics of Information. *Journal of Political Economy* 69(3). · Coase, R. H. (1937). The Nature of the Firm. *Economica* 4(16).
12. Myerson, R. B. & Satterthwaite, M. A. (1983). Efficient Mechanisms for Bilateral Trading. *Journal of Economic Theory* 29(2). · Chatterjee, K. & Samuelson, W. (1983). Bargaining under Incomplete Information. *Operations Research* 31(5).
13. Backus, M., Blake, T. & Tadelis, S. (2019). On the Empirical Content of Cheap-Talk Signaling: An Application to Bargaining. *Journal of Political Economy* 127(4).
14. Einav, L., Farronato, C. & Levin, J. (2016). Peer-to-Peer Markets. *Annual Review of Economics* 8. · Roth, A. E. (2008). What Have We Learned from Market Design? *The Economic Journal* 118(527).
15. Bianchi, F. et al. (2024). How Well Can LLMs Negotiate? NegotiationArena Platform and Analysis. ICML 2024.
16. Bansal, G. et al. (2025). Magentic Marketplace: An Open-Source Environment for Studying Agentic Markets. Microsoft Research.
17. Pew Research Center (2021). The State of Gig Work in 2021.
18. Chen, M. K., Rossi, P. E., Chevalier, J. A. & Oehlsen, E. (2019). The Value of Flexible Work: Evidence from Uber Drivers. *Journal of Political Economy* 127(6).
19. Horton, J. J. & Zeckhauser, R. J. (2016). Owning, Using and Renting: Some Simple Economics of the "Sharing Economy". NBER w22029.
20. Farronato, C. & Fradkin, A. (2022). The Welfare Effects of Peer Entry: The Case of Airbnb and the Accommodation Industry. *American Economic Review* 112(6).
21. ThredUp 2025 Resale Report, via Chain Store Age and Retail Dive.
22. National Student Clearinghouse Research Center, Fall 2025 enrollment.
23. UW News, Enrollment increases across all three UW campuses, Oct 30, 2025.
24. Hadfield, G. K. & Koh, A. (2025). An Economy of AI Agents. NBER Handbook on the Economics of Transformative AI.
25. Further reading: Resnick & Zeckhauser (2002) on eBay reputation; Tadelis (2016) on platform feedback; Katz & Krueger (2019) on alternative work; Cohen et al. (2016) on consumer surplus from Uber; Castellani, Sala & Mirabella (2015) on the environmental benefit of reuse.